NAU
NORTHERN ARIZONA UNIVERSITY

Hydro-Power Collegiate Competition(HCC) Capstone

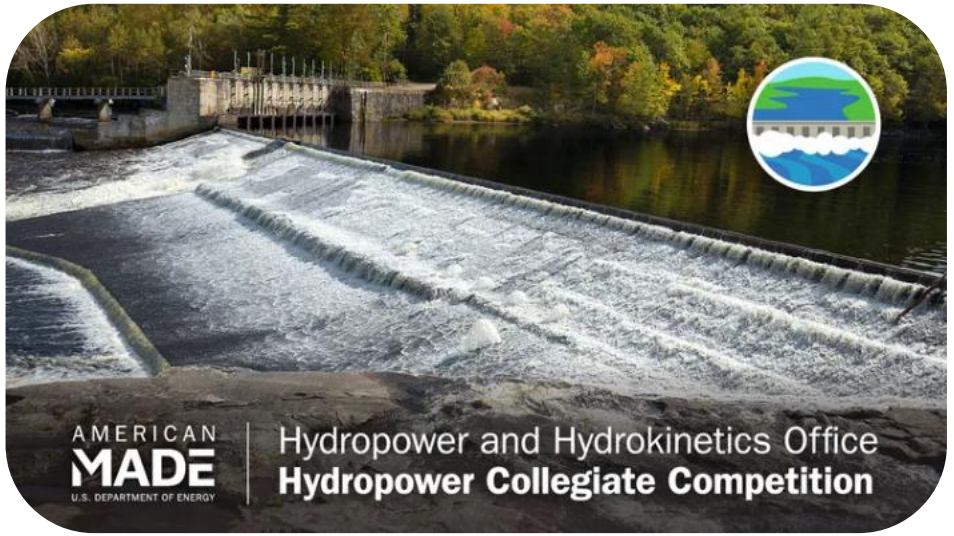
Tyler Loggins

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Zachery Mills

Robert Wanek

Ben Krotonsky



Project Description



Select and evaluate a U.S. non-powered dam (NPD) and develop a feasible hydropower retrofit that increases renewable-energy generation while maintaining the dam's existing function.



Advisor / Client:
Carson Pete

Sponsor:
NAU Energy Club

Sponsor:
U.S. Department of Energy

Why this Project Matters

- Less than 5% of dams in the United States are used to generate power
- Transforms existing infrastructure into usable power
- Supports the environment by utilizing renewable energy
- Real life examples of the benefits of hydropower
- Builds experience for the next generation of hydropower engineers
- Provides the team with hand on experience tackling real life engineering problems



Benchmarking

HCC 2024
Non-Power Dam



Kentucky River Dam
1st Place Overall

HCC 2025
Closed Loop Pump



Figure 20: Fluid Lab Francis

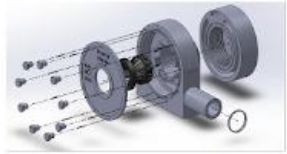


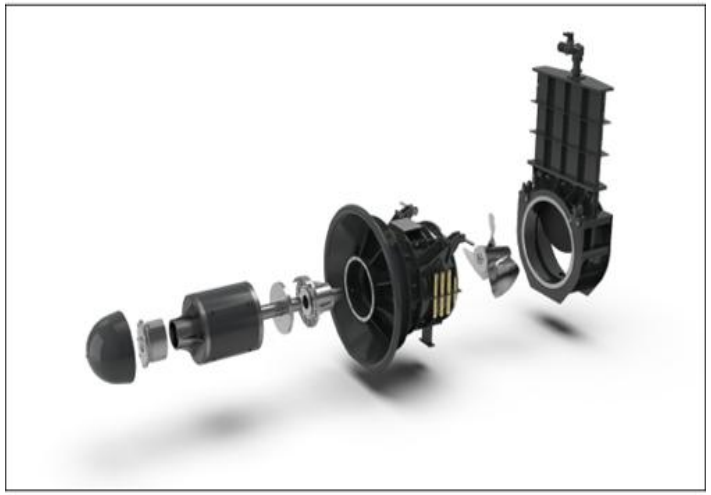
Figure 21: Francis CAD Model



Figure 22: Assembled Turbine

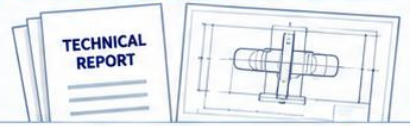
Design a Closed-Loop PSH System
Up to 1 GW Storage
8-24hr Storage

HCC 2026
Hydro-Power Innovation



Modular submerged turbines
Siting Competition Winner
3rd Place Overall

Customer Requirements

#	Customer Requirement	Image	Description
CR1	Siting		Select a viable non-powered dam site.
CR2	Feasibility		Demonstrate that the selected site can realistically support hydropower.
CR3	Design		Develop an engineering design for the selected site.
CR4	Innovation		Provide a new or improved hydropower solution.
CR5	Engagement		Connect with industry and community stakeholders.
CR6	Documentation		Clearly document the siting and design work.
CR7	Presentation		Communicate the project and results effectively.
CR8	Prototyping		Build and test a scaled prototype of the concept.

Engineering Requirements

#	Engineering Requirement	Image	Description	Units	Goal (Target Value)
ER1	**Power Output**		Generate a target average power output at the selected site.	kW	**≥ 500**
ER2	**Efficiency**		Achieve high overall system efficiency (turbine, generator, and balance of plant).	%	**≥ 80**
ER3	**Structural Integrity**		Ensure components can withstand hydraulic, mechanical, and environmental loads.	Safety Factor (FOS)	**≥ 2.0**
ER4	**Environmental Impact**		Minimize impacts to aquatic life, water quality, and the surrounding ecosystem.	Qualitative / Index	**Low Impact** (meet or exceed guidelines)
ER5	**Cost Effectiveness**		Deliver a cost-effective solution with reasonable installation, operation, and maintenance costs.	$/kW	**≤ 2,000**
ER6	**Reliability**		Achieve reliable operation with minimal downtime and low maintenance requirements.	Availability (%)	**≥ 95**
ER7	**Scalability**		Design a modular solution that can be scaled to different sites and flow conditions.	Module Size (kW/unit)	**100 – 500** (per unit)
ER8	**Constructability**		Enable practical construction, installation, and integration at the selected site.	Installation Time (months)	**≤ 12**

QFD

Customer Requirements (WHATs)		Importance (1–5)	Engineering Requirements (HOWs)								Total (Importance Weighted)
			ER1 Power Output	ER2 Efficiency	ER3 Structural Integrity	ER4 Environmental Impact	ER5 Cost Effectiveness	ER6 Reliability	ER7 Scalability	ER8 Constructability	
CR1	Siting	5	9	3	3	9	9	1	1	9	175
CR2	Feasibility	5	9	9	9	9	9	9	3	9	275
CR3	Design	5	9	9	9	3	3	9	3	9	225
CR4	Innovation	4	3	9	1	3	3	3	9	3	136
CR5	Engagement	3	1	0	0	9	3	1	1	3	54
CR6	Documentation	2	1	1	1	1	1	1	1	1	16
CR7	Presentation	2	1	1	1	1	1	1	1	1	16
CR8	Prototyping	4	9	9	9	1	3	9	3	9	172
Technical Requirement Score (Sum of weighted relationships)			168	162	154	148	138	138	90	172	1,170
Normalized Weight (% of total)			14.4%	13.8%	13.2%	12.6%	11.8%	11.8%	7.7%	14.7%	100%

Relationship Strength:	9 = Strong	3 = Moderate	1 = Weak	0 = None

Literature Review

U.S. DOE / HeroX – 2027 Hydropower Collegiate Competition [1]

Topic: Project / HCC Requirements

FERC – Small/Low-Impact Hydropower Project Guidance [2]

Topic: Codes / Regulations

U.S. Bureau of Reclamation – Commissioning Guide for Hydroelectric Facilities, FIST 4-7 [3]

Topic: Standards / Prototype Testing

U.S. Bureau of Reclamation – Performance Testing Multiple Units of Similar Hydraulic Design [4]

Topic: Standards / Turbine Performance

Hansen et al. – Technical Hydropower Potential at U.S. Non-Powered Dams [5]

Topic: Non-Powered Dams / Mathematical Modeling

White & Xue – Fluid Mechanics [6]

Topic: Fluid Mechanics / Mathematical Modeling

Sasthav & Oladosu – Environmental Design of Low-Head Run-of-River Hydropower in the United States [7]

Topic: Environmental Impacts

Literature Review

Efficient hydro-energy conversion technologies, challenges, and policy implication [8]

Topic: Turbine Selection / Head and Flow Rate / Seasonal Flow

Short-cut design of small hydroelectric plants [9]

Topic: Flow Rate / Turbine Design / Economic Considerations

Opportunities for Energy Development in Water Conduits [10]

Topic: DOE / In-Conduit Hydropower

Recovering energy by hydro-turbines application in water transmission pipelines: A case study west of Saudi Arabia [11]

Topic: Pressurized Water Lines

An Assessment of Technical Hydropower Potential at Non-Powered Dams in the United Stutes [2]

Topic: NPD Site Considerations

Contextualizing Non-Powered Dam Site Selection for Archimedes Screw Turbines: A Methodology for Responsible Archimedes Screw Turbine Conversion at Existing Dams [12]

Topic: Archimedes Screw Turbines / Fish Passage

An Assessment of Hydropower Potential at National Conduits [13]

Topic: In-Conduit Turbines / Potential Locations

Water Data For the Nation [14]

Topic: National Water Data / Flow Rates

Literature Review



Literature Review



Literature Review

California's In-Conduit Hydropower Implementation Guidebook [29]
Topic: In-Conduit Hydropower / Implementation Guidance

Design Models for Small Run-of-River Hydropower Plants [30]
Topic: Run-of-River Hydropower / Modeling and Design

Hydropower Development Potential at Non-Powered Dams [31]
Topic: Development Potential / Data Needs

National Inventory of Dams [32]
Topic: Dam Data / National Inventory

Phase 2: Site Selection and Feasibility Assessment (ORNL) [33]
Topic: Site Selection / Feasibility Assessment

Phase 3: Design and Technology Selection (ORNL) [34]
Topic: Design and Technology Selection

StreamStats | U.S. Geological Survey [35]
Topic: Hydrologic Data / Streamflow Analysis

Mathematical Modelling

Design Question

How does penstock diameter affect water velocity and hydraulic head loss and what diameter should be carried forward for preliminary design?

Engineering Tools & References

Microsoft Excel
Parametric calculations and data analysis

MATLAB
Model multiple design scenarios and optimize diameter

White & Xue – Fluid Mechanics
Validate governing equations (continuity, head loss)

ASCE MOP 79 – Steel Penstocks
Check design against accepted penstock design practices

Governing Equations

Concept	Equation	Meaning
Area	$A = \frac{\pi D^2}{4}$	Cross-sectional area of penstock
Velocity	$V = \frac{Q}{A}$	Average water velocity in penstock
Friction loss (Darcy-Weisbach)	$h_f = f \frac{L}{D} \frac{V^2}{2g}$	Head loss due to pipe friction
Minor losses	$h_m = \sum K \frac{V^2}{2g}$	Head loss from fittings, entrances, bends, etc.
Total head loss	$h_L = h_f + h_m$	Total hydraulic losses in penstock

System Diagram



Mathematical Modelling

Assumed Parameters

Parameter	Symbol	Value	Units
Flow rate	Q	5.0	m^3/s
Penstock length	L	75	m
Friction factor	f	0.020	–
Sum of minor loss coeff.	ΣK	2.5	–

Design Impact & Next Steps

- 1.50 m penstock selected as preliminary design diameter.
- Reduces hydraulic losses significantly while avoiding unnecessary increase in pipe size and cost.
- Next steps: refine using site-specific flow data, evaluate structural stress and surge (water hammer), and assess cost trade-offs.

Validation

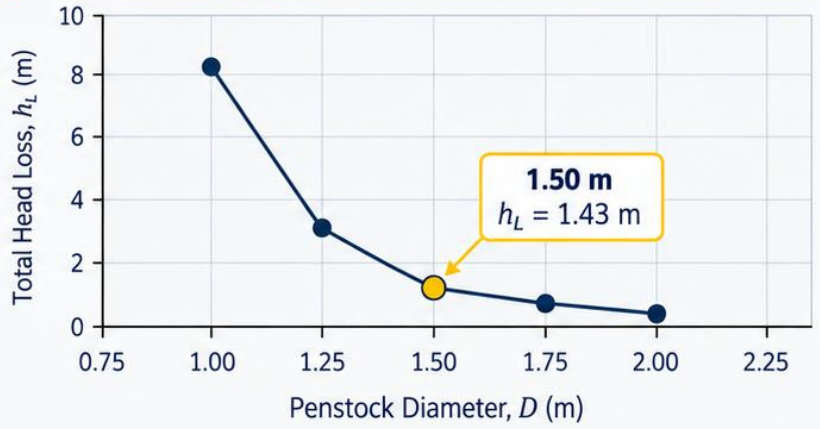
- Calculations follow continuity and Darcy-Weisbach methods from White & Xue.
- Results were independently reproduced in Excel and MATLAB.
- Trends match expected behavior: larger diameter → lower velocity → lower losses.

Calculation Results

Diameter (m)	Area (m^2)	Velocity (m/s)	Friction Loss (m)	Minor Loss (m)	Total Loss (m)
1.00	0.785	6.37	3.10	5.16	8.26
1.25	1.227	4.07	1.02	2.12	3.13
1.50	**1.767**	**2.83**	**0.41**	**1.02**	**1.43**
1.75	2.405	2.08	0.19	0.55	0.74
2.00	3.142	1.59	0.10	0.32	0.42

Increasing the penstock diameter from 1.25 m to 1.50 m reduces total hydraulic losses by **54%** (3.13 m → 1.43 m).

Head Loss vs. Diameter



Mathematical Modelling

Design Question

How can site flow-duration data and turbine type be used to estimate an optimal nominal design flow?

Governing Equation

$$Q_n = \left[\frac{\gamma q_{50}}{1 + (\gamma - 1)q_{50}} * \left(1 - \frac{q_{min}}{q_{max}}\right) + \frac{q_{min}}{q_{max}}\right] Q_{max}$$

Parameter Definitions

Parameter	Definition
Q_n	nominal flow rate
q_{50}	site median flow
q_{max} and q_{min}	site max and min flow
Q_{max}	site maximum flow rate
γ	turbine coefficient

Turbine Coefficient, γ

Turbine	γ
Francis	0.422
Pelton	0.369
Axial	0.364

Key Takeaways

- Selects an optimal design flow using site flow data and turbine type.
- Helps balance energy capture and project cost.
- Can be applied to candidate sites and turbine options for comparison.

Mathematical Modelling

Governing Equation

$$Q_n = \left[\frac{\gamma q_{50}}{1+(\gamma-1)q_{50}} * \left(1-\frac{q_{min}}{q_{max}}\right)+\frac{q_{min}}{q_{max}}\right] Q_{max}$$

Validation Example

Verde River example used to validate the optimum-flow equation.

Key Takeaways

- Verde River example gives $Q_n = 253.5\ ft^3/s$ for a Francis turbine.
- The method uses flow-duration data to select design flow.
- Helps balance energy capture and project cost.
- Can be applied to candidate sites and turbine options for comparison.

Calculations

Parameter	Calculation / Value	Result
Maximum flow rate (at site)	$Q_{max} = 1367.6\,ft^3/s$	$1367.6\,ft^3/s$
Normalized median flow	$q_{50} = \frac{Q_{50}}{Q_{max}} = \frac{268.9}{1367.6}$	0.197
Normalized minimum flow	$q_{min} = \frac{Q_{min}}{Q_{max}} = \frac{138.4}{1367.6}$	0.101
Normalized maximum flow	$q_{max} = 1$	1
Turbine coefficient (Francis)	$\gamma = 0.422$ (Francis)	0.422
Optimum design flow	$Q_n = 253.5\ ft^3/s$	**253.5 ft³/s**

Mathematical Modelling

Design Question: On the lower end of power production for this competition what is a realistic size of the shaft transfering power from the turbine to the generator?

Purpose

MATLAB will be used to predict shaft stress and fatigue while transferring power from the turbine to the generator.

System Flow

Turbine Power, P → **Shaft** Torque, T → **Stress** τ, σ → **Fatigue** Safety check → **Generator** Output

Key Equations

1 Torque from Power

$$T = P / \omega$$

$\omega = 2\pi n / 60$

$T = 9550\ P(kW) / n(rpm)$

2 Torsional Shear Stress

$$\tau = Tr / J$$

3 Solid-Shaft Form

$$\tau_{max} = 16T / (\pi d^3)$$

4 Angle of Twist

$$\theta = TL / (JG)$$

Combined Loading & Fatigue

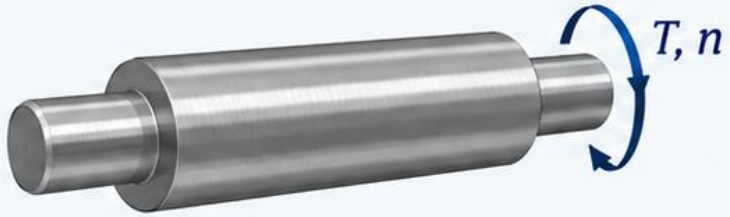
Bending: $\sigma = 32M / (\pi d^3)$ | **Von Mises:** $\sigma_{VM} = \sqrt{(\sigma^2 + 3\tau^2)}$ | **Fatigue:** evaluate alternating + mean stress

MATLAB will compare shaft diameters and materials against stress, fatigue, and factor-of-safety limits.

Mathematical Modelling

Design Question

Size a solid shaft for 1 MW at 300 rpm with τ_{allow} = 50 MPa.



Inputs

- $P = 1$ MW
- $n = 300$ rpm
- $\tau_{allow} = 50$ MPa

Equations

$$T = 9550 \frac{P}{n}$$

$$d = \left(\frac{16T}{\pi \tau_{allow}} \right)^{1/3}$$

Results

$$T = 31.83 \text{ kN·m}$$

$$d \approx 148 \text{ mm} = 5.82 \text{ in}$$

Next Step

Refine using bending, fatigue, keyways, stress concentrations, and MATLAB analysis.

Mathematical Modelling

Can estimate potential hydraulic power of existing non powered dams.

$$P_h = \rho g Q H_{net} \eta$$

Variable		Dependent?	Value	Source
ρ	Fluid Density	N	Water – 1000 $\frac{kg}{m^3}$. (STP)	—
g	Gravity Constant	N	9.81 $\frac{m}{s^2}$	—
Q	Volumetric Flow Rate	Y	—	Data or $V_{flow} \times A_{enter}$
H_{net}	Total Head	Y	—	Hydraulic Height Data
η	Efficiency	Y	—	Site & Turbine Combination

Data Point	Source	Utilization
Q, Volumetric Flow Rate	USGS Flow Statistics	Apply directly from Q30 data points. Compare gauge locations to dam source and proximity.
H_{net}, Net Head	National Inventory of Dams (NID)	Apply directly from hydraulic height data points.
η, Efficiency	Standard Power Estimation Assumption	Apply similar efficiency coefficients to replicate successful site selection.

This is the methodology used by Oak Ridge National Laboratory to produce energy estimates and analyze potential NPD sites.

Mathematical Modelling

$$P_h = \rho g Q H_{net} \eta$$

Variable	Value
Q – Volumetric Flow Rate	19.10 $(\frac{m^3}{s})$
H_{net} - Total Head	32.92 (m)
η – Hydroelectric Efficiency	85%

$$P_h = (1000)(9.81)(19.10)(32.92)(.85)$$

$$P_h = 5.25\ MW$$

This is the methodology used by Oak Ridge National Laboratory to produce energy estimates and analyze potential NPD sites.

Mathematical Modelling

- **Ranks NPDs from Oak Ridge National Laboratory inventory [5] based on expected yearly power output and flow condition variation between months**
- **Ranking the NPDs will allow the team to decide on siting with numbers to defend it**
- **Excluded if theoretical instantaneous power capacity, $E_{instant}$, was outside of the bounds of competition requirements (1-10 MW)**

Metric	Optimal Condition	Scoring	Weight
Theoretical Yearly Power Generation	Highest yearly output (MWh)	$S_{energy} = {E_{annual} - E_{min}}/{E_{max} - E_{min}}$	50%
Flow Variance	Lowest flow rate variance month-to-month	$S_{flow} = {CV_{flow,max} - CV_{flow}}/{CV_{flow,max} - CV_{flow,min}}$	25%
Head Variance	Lowest head variance month-to-month	$S_{head} = {CV_{head,max} - CV_{head}}/{CV_{head,max} - CV_{head,min}}$	25%

$$CV = \frac{\sigma_{month}}{\mu_{month}}$$

Mathematical Modelling

rank	id	name	st	river	capMW_instant	MWh/yr	en_Scr	fl_CV	fl_Scr	hd_CV	hd_Scr	score
1	{'NM00404'}	"COCHITI DAM "	{'NM'}	"Rio Grande & Sa "	9.91	58407	1	0.61	0.49	0.02	0.96	0.86
2	{'WA00769'}	"FISH BARRIER DA "	{'WA'}	"Baker River "	7.88	40651	0.68	0.25	0.8	0.07	0.85	0.76
3	{'AZ10308'}	"BARTLETT DAM "	{'AZ'}	"Verde River "	8.26	34442	0.57	0.32	0.74	0.04	0.92	0.7
4	{'WA00555'}	"BARRIER DAM "	{'WA'}	"Cowlitz River "	6.23	36358	0.61	0.38	0.69	0.07	0.85	0.69
5	{'CA00226'}	"ANDERSON COTTON "	{'CA'}	"Sacramento Reve "	7.44	39618	0.66	0.42	0.66	0.13	0.72	0.68
6	{'CA00034'}	"FISH BARRIER "	{'CA'}	"Feather River "	2.56	19868	0.31	0.02	1	0.04	0.92	0.64
7	{'KY03009'}	"BARREN RIVER DA "	{'KY'}	"Barren River "	7.93	33459	0.55	0.34	0.72	0.15	0.66	0.62
8	{'IL01014'}	"PEORIA LOCK AND "	{'IL'}	"Illinois River "	9.7	39929	0.67	0.39	0.68	0.24	0.47	0.62
9	{'KS00012'}	"TUTTLE CREEK DA "	{'KS'}	"Big Blue River "	9.49	32058	0.53	0.59	0.51	0.04	0.92	0.62
10	{'IL00115'}	"JERRY F. COSTEL "	{'IL'}	"Kaskaskia River "	9.13	34938	0.58	0.64	0.47	0.08	0.83	0.61

= Optimal = Sub-Optimal

Min:	2411 MWh	0.02	0
Max:	58407 MWh	1.19	0.44

Example Score Assessment – Cochiti Dam

$$S_{tot} = 0.5 \cdot S_{energy} + 0.25 \cdot S_{flow} + 0.25 \cdot S_{head}$$

$$\Rightarrow S_{tot} = 0.5 \cdot \frac{58407 - 2411}{58407 - 2411} + 0.25 \cdot \frac{1.19 - 0.61}{1.19 - 0.02} + 0.25 \cdot \frac{0.44 - 0.02}{0.44 - 0}$$

$$\Rightarrow S_{tot} = 0.5 \cdot 1 + 0.25 \cdot 0.496 + 0.25 \cdot 0.955$$

$$\Rightarrow S_{tot} = 0.5 + 0.124 + 0.239 = 0.86$$

Mathematical Modelling

rank	id	name	st	river	capMW_instant	MWh/yr	en_Scr	fl_CV	fl_Scr	hd_CV	hd_Scr	score
1	{'NM00404'}	"COCHITI DAM "	{'NM'}	"Rio Grande & Sa "	9.91	58407	1	0.61	0.49	0.02	0.96	0.86
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Optimal NPD Selection:

Cochiti Dam, NM

NPD Candidate Capacity Ranges

Capacity range	Count	% of dataset
< 1 MW	2,320	88.7%
1–10 MW (competition)	228	8.7%
> 10 MW	68	2.6%

Budget

Team Budget

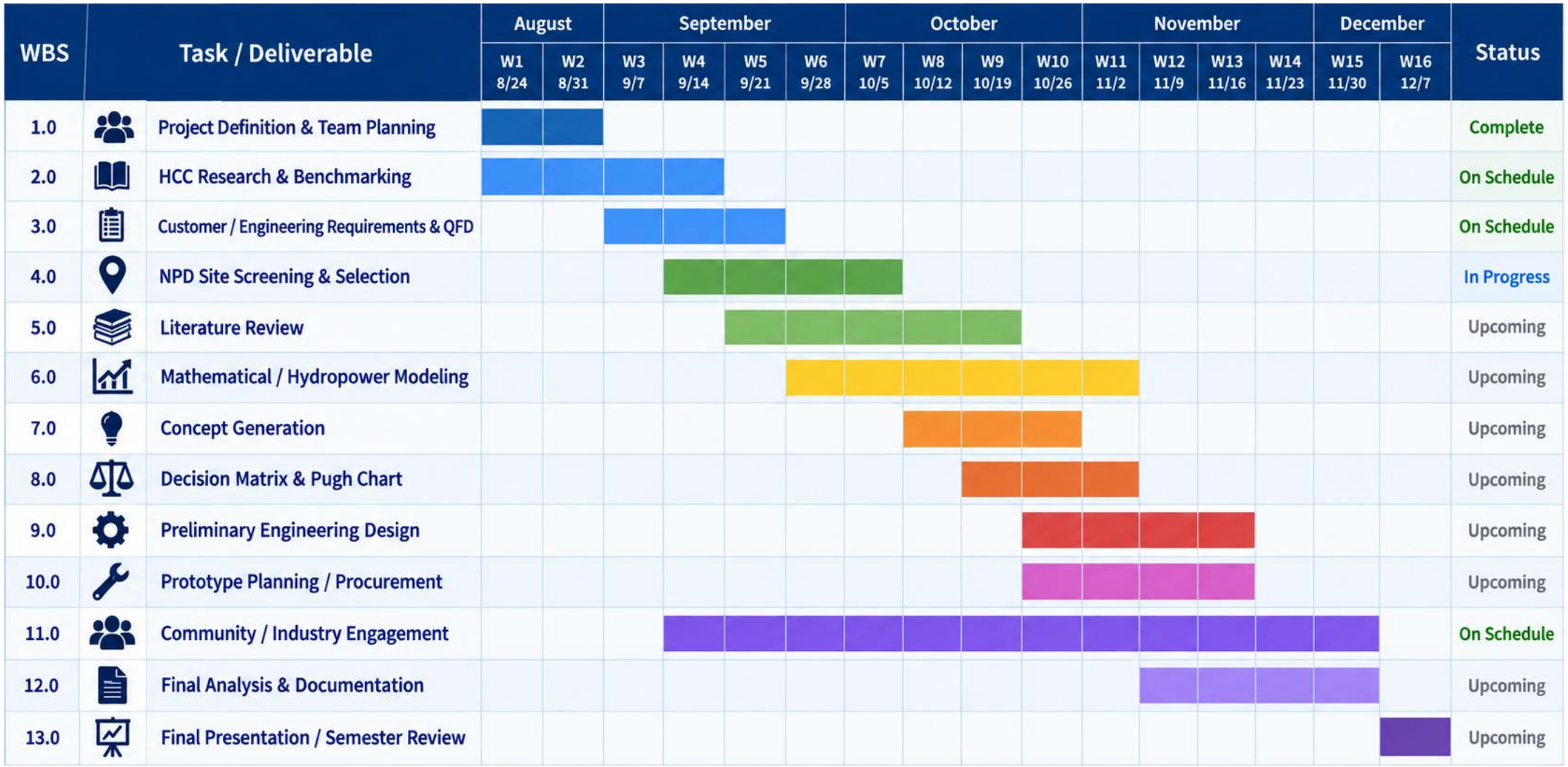
Category	Image	Budget
Prototype Components		$4,000
Fabrication / Materials		$2,000
Testing & Instrumentation		$2,000
Travel / Competition		$3,000
Community Engagement		$750
Poster / Presentation		$500
Site Research / Data		$750
Contingency		$2,000
Total		**$15,000**

Minimum Fundraising Goal (10%) = $1,500

Theoretical Project Budget

Category	Image	Estimated Allocation
Turbine & Generator		20%
Civil / Structural Modifications		30%
Electrical & Grid Interconnection		15%
Mechanical / Balance of Plant		10%
Engineering & Design		7%
Environmental / Permitting		3%
Construction / Mobilization		5%
Contingency		10%
Total		**100%**

Schedule



WBS	Task / Deliverable	August		September				October				November				December		Status
		W1 8/24	W2 8/31	W3 9/7	W4 9/14	W5 9/21	W6 9/28	W7 10/5	W8 10/12	W9 10/19	W10 10/26	W11 11/2	W12 11/9	W13 11/16	W14 11/23	W15 11/30	W16 12/7	
1.0	Project Definition & Team Planning																	Complete
2.0	HCC Research & Benchmarking																	On Schedule
3.0	Customer / Engineering Requirements & QFD																	On Schedule
4.0	NPD Site Screening & Selection																	In Progress
5.0	Literature Review																	Upcoming
6.0	Mathematical / Hydropower Modeling																	Upcoming
7.0	Concept Generation																	Upcoming
8.0	Decision Matrix & Pugh Chart																	Upcoming
9.0	Preliminary Engineering Design																	Upcoming
10.0	Prototype Planning / Procurement																	Upcoming
11.0	Community / Industry Engagement																	On Schedule
12.0	Final Analysis & Documentation																	Upcoming
13.0	Final Presentation / Semester Review																	Upcoming

References

[1] “2027 Hydropower Collegiate Competition | HeroX.” Accessed: Sep. 15, 2026. [Online]. Available: https://www.herox.com/hydropower-collegiate-competition-2027

[2] “An Assessment of Technical Hydropower Potential at Non-Powered Dams in the United States.” Accessed: Sep. 15, 2026. [Online]. Available: https://hydrosource.ornl.gov/publications/an-assessment-of-technical-hydropower-potential-at-non-powered-dams-in-the-united-states/

[3] “Small/Low Impact Hydropower Projects | Federal Energy Regulatory Commission.” Accessed: Sep. 15, 2026. [Online]. Available: https://www.ferc.gov/licensing/smalllow-impact-hydropower-projects

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[6] “ISE Fluid Mechanics.” Accessed: Sep. 15, 2026. [Online]. Available: https://www.mheducation.co.uk/ise-fluid-mechanics-9781260575545-emea-group

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Thank You
& Questions?